

JUNDA ZHAO

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Research Interests

Reliable testing and verification for AI coding agents, with an emphasis on establishing the intended behavior of code when the implementation itself may be incorrect, generating tests that reveal defects rather than encode them, and assessing whether the evidence used to validate AI-generated software can be trusted. **Seeking a postdoctoral position starting in late 2027.**

Education

Doctor of Philosophy (Ph.D. Candidate), Mechanical and Industrial Engineering 09/2022 to present

University of Toronto, St. George Campus, Toronto, Canada

- Advisor: Prof. Eldan Cohen, Optimization and Machine Learning (OptiMaL) Lab; close collaboration with Prof. Shurui Zhou (Electrical and Computer Engineering)
- Dissertation (in progress): *Accurate Specification Recovery for Effective Test Generation*
- Passed the doctoral qualifying examination; expected completion in 2027

Honours Bachelor of Science, Computer Science 09/2016 to 11/2021

University of Toronto, St. George Campus, Toronto, Canada

Selected coursework

- Graduate: CSC2511 Natural Language Computing (A+); MAT1510 Deep Learning: Theory and Data Science (A+); MIE1513 Decision Support Systems (A+)
- Undergraduate: CSC412 Probabilistic Learning and Reasoning (A+/94); CSC413 Neural Networks and Deep Learning (A+/99); CSC420 Introduction to Image Understanding (A+/96)

Honours and Awards

ACM SIGSOFT Distinguished Paper Award, ISSTA 2026 2026

For “Do Coverage and Mutation Scores of LLM-Generated Test Suites Correlate with Their Effectiveness? (Replicability Study)”

Ontario Graduate Scholarship (OGS), Government of Ontario and University of Toronto

2024 to 2025

Publications

- [1] **Junda Zhao**, Shurui Zhou, Eldan Cohen. “Do Coverage and Mutation Scores of LLM-Generated Test Suites Correlate with Their Effectiveness? (Replicability Study).” *ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2026)*. **ACM SIGSOFT Distinguished Paper Award**. [arXiv](#)
- [2] **Junda Zhao**, Shurui Zhou, Eldan Cohen. “Evaluating and Mitigating the Misguidance Effect of Buggy Code in LLM-Generated Unit Tests.” *ACM SIGSOFT International Symposium on Software Testing and Analysis (ISSTA 2026)*. [arXiv](#)
- [3] **Junda Zhao**, Yuliang Song, Eldan Cohen. “Variational Prefix Tuning for Diverse and Accurate Code Summarization Using Pre-trained Language Models.” *Journal of Systems and Software*, 2025. [Paper](#)
- [4] Faria Khandaker, Arik Senderovich, **Junda Zhao**, Eldan Cohen, Eric Yu, Sebastian Carbajales, Allen Chan. “Transformer Models for Mining Intents and Predicting Activities from Emails in Knowledge-Intensive Processes.” *Engineering Applications of Artificial Intelligence*, 2024. [Paper](#)
- [5] **Junda Zhao**, Weiling Cai, Yizhen Ge, Lijuan Zhang. “A New Fuzzy Belonging-based Multi-view K-means Clustering Algorithm.” *IEEE Annual Information Technology, Electronics and Mobile Communication Conference (IEMCON 2019)*. [Paper](#)
- [6] **Junda Zhao**. “Accurate Specification Recovery for Effective Test Generation.” *SPLASH/ISSTA 2026 Doctoral Symposium*.

Research Experience

Ph.D. Candidate, OptiMaL Lab, Department of Mechanical and Industrial Engineering

09/2022 to present

University of Toronto, supervised by Prof. Eldan Cohen, Toronto, Canada

- Reliable testing and verification for AI coding agents, with a focus on LLM-based unit test generation.
- Specification-based testing: recovering the intended behavior of code that may be buggy, and generating tests from it.
- Evaluation of LLM-generated tests and of the metrics used to judge them.
- Improving the reliability of testing and verification when people work with AI-assisted development tools.
- Code summarization with pre-trained language models of code.

Research Assistant, Dynamic Graphics Project (DGP) Lab

11/2021 to 09/2022

University of Toronto, Toronto, Canada

- Research on geometric stylization: developed a method for texture synthesis and transfer based on a feature-grid neural implicit representation of arbitrary objects, whose 2-D structure allows computer vision and 2-D geometry processing techniques to be applied to 3-D representations.

Undergraduate Projects, Department of Computer Science, University of Toronto 10/2020 to 09/2021

- **Human musculoskeletal system simulation** (CSC494 research project): proposed, built, and simulated a fundamental unit of the musculoskeletal system from mass points, rods, and springs, with machine learning for fast approximation of the system state.
- **Accurate large-scale ferrofluid simulation** (CSC417 team course project): implemented the ferrofluid simulation of Huang et al. and studied the formation of spikes under a magnetic field.

Talks and Posters

Research-track talk and poster, ISSTA 2026, Oakland, CA, USA 10/2026

“Do Coverage and Mutation Scores of LLM-Generated Test Suites Correlate with Their Effectiveness? (Replicability Study)”

Research-track talk and poster, ISSTA 2026, Oakland, CA, USA 10/2026

“Evaluating and Mitigating the Misguidance Effect of Buggy Code in LLM-Generated Unit Tests”

Journal-First talk and poster, ASE 2025, Seoul, South Korea 11/2025

“Variational Prefix Tuning for Diverse and Accurate Code Summarization Using Pre-trained Language Models”

Doctoral Symposium talk, SPLASH/ISSTA 2026, Oakland, CA, USA 10/2026

“Accurate Specification Recovery for Effective Test Generation”

Teaching Experience

Teaching Assistant, MIE1517 Introduction to Deep Learning (graduate course) 01/2024 to 04/2024; 09/2024 to present

University of Toronto, Department of Mechanical and Industrial Engineering

- Fall and Winter terms since 09/2024. Lead tutorial sessions, help develop and improve course materials and plans, and grade projects and tests.

Teaching Assistant, MIE370 Introduction to Machine Learning (undergraduate course) 09/2025 to 12/2025

University of Toronto, Department of Mechanical and Industrial Engineering

- Led tutorial sessions, helped prepare teaching materials, and graded projects and tests.

Teaching Assistant, MIE350 Design and Analysis of Information Systems (undergraduate course) 09/2022 to 12/2022

University of Toronto, Department of Mechanical and Industrial Engineering

- Led tutorial sessions, helped prepare teaching materials, and graded projects and tests.

Professional Experience

Junior Identity and Access Management Analyst 09/2019 to 09/2020

University of Toronto, Information Technology Services, Toronto, Canada

- Built and maintained a database system for tracking and analyzing network access and activity information for students, faculty, and staff on the university network.